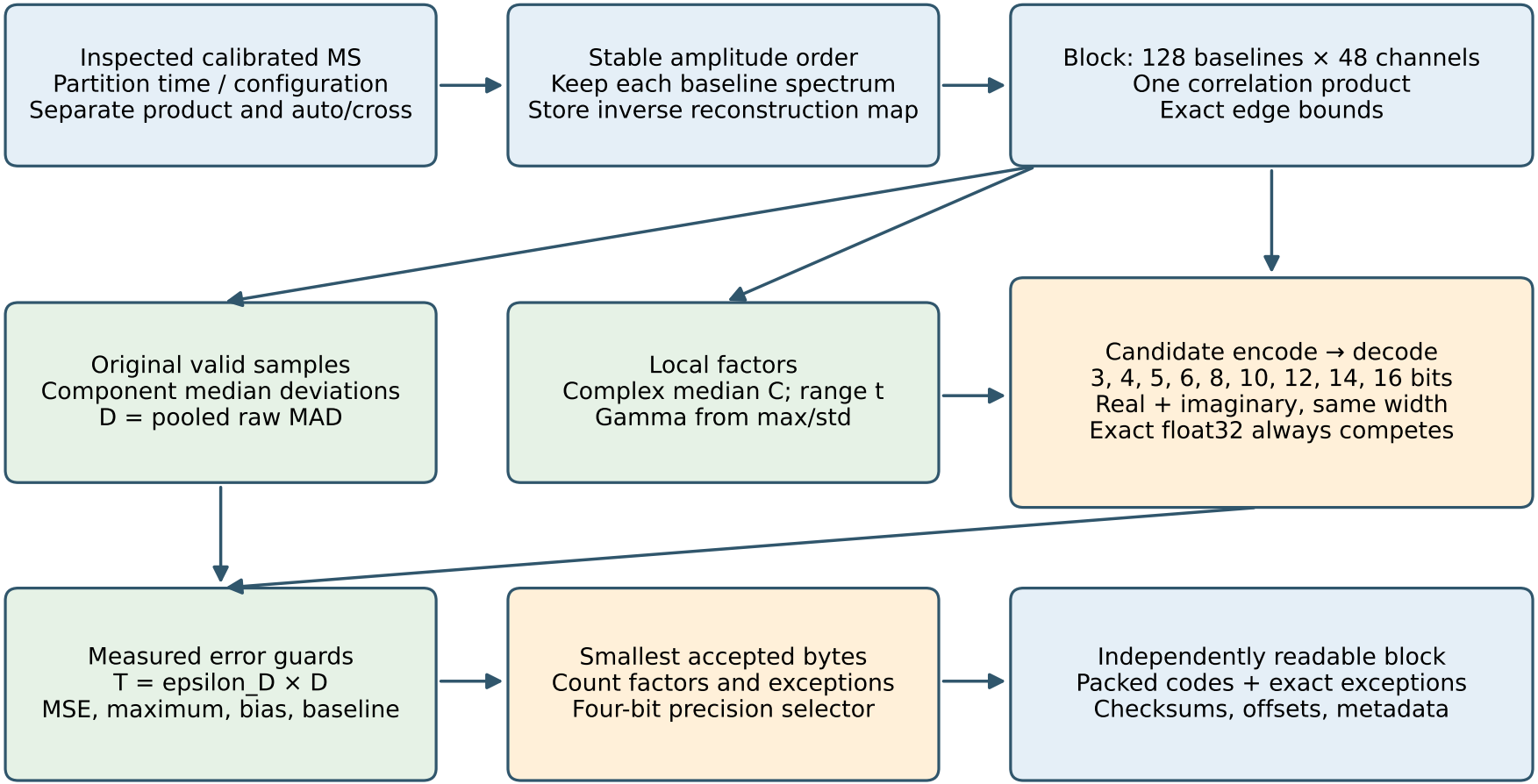Grouping, local factors and adaptive precision